CAPSTONE PROJECT

Custom ALV Report for Vendor Invoice Analysis

SAP ABAP | Program: **Z_VENDOR_INVOICE_REPORT**

Student Name	PRAKHAR WADHWANI
Roll Number	23053143
Branch	BTECH/CSE
Submission Date	April 21, 2026
Program ID	Z_VENDOR_INVOICE_REPORT

1. Title & Problem Statement

Project Title: Custom ALV Report for Vendor Invoice Analysis in SAP ABAP

In any SAP-based enterprise, the Accounts Payable (AP) department processes hundreds of vendor invoices every day. Finance teams require a consolidated, realtime view of both *open* (unpaid) and *cleared* (paid) vendor invoices — including invoice amounts, posting dates, payment terms, and clearing details — to manage cash flow and vendor relationships effectively.

SAP does not provide a single standard report that combines open items (BSIK) and cleared items (BSAK) with vendor master data (LFA1) in a flexible, ALV-based output. Finance controllers must individually check transactions FK10N, FBL1N, and ME23N, causing manual effort and delays.

Problem: There is no unified, filterable, and exportable vendor invoice report in the standard SAP system that covers both open and cleared invoices with vendor names, payment status, and subtotals per vendor — accessible to business users without technical knowledge.

A custom ABAP report **Z_VENDOR_INVOICE_REPORT** was developed using the

Event / Form	Purpose
INITIALIZATION	Sets screen title text symbols

2. Solution & Key Features

REUSE_ALV_GRID_DISPLAY function module. The program fetches data from SAP standard tables, enriches it with vendor master information, and presents a consolidated, interactive ALV grid output.

Feature	Description
Selection Screen	Flexible filters: Company Code, Vendor, Date range, Fiscal Year, Document T
Open + Cleared Items	Checkboxes to include BSIK (open) and BSAK (cleared) items independently
Vendor Name Enrichment	Joins LFA1 master table to display Vendor Name alongside Vendor Number
ALV Grid Output	Sortable, filterable, color-striped ALV grid with column-width optimization
Subtotals & Grand Total	Automatic subtotals per Company Code and Vendor; Grand Total row at botto
Payment Status Column	Clearly labels each row as "Open" or "Cleared" for instant visibility
Double-Click Navigation	Double-clicking any row opens the FI document directly in FB03
Display Variant	Users can save and reload personalized column layouts via ALV variants
Export Ready	Built-in ALV toolbar allows export to Excel, PDF, and print

3. Tech Stack

Component	Details
Language	ABAP (Advanced Business Application Programming)
SAP Platform	SAP ECC 6.0 / SAP S/4HANA
Key Tables	BSIK (AP Open Items), BSAK (AP Cleared Items), LFA1 (Vendor Mast
ALV Function	REUSE_ALV_GRID_DISPLAY (SAP List Viewer)
Include Library	SLIS (SAP List Viewer Utilities)
Transaction Used	SE38 (ABAP Editor), FB03 (FI Document Display)
SAP Module	FI-AP (Financial Accounting – Accounts Payable)
Development Class	ZFIN (Custom Finance Development)

4. ABAP Code Walkthrough & Screenshots

4.1 Program Architecture

The program follows a modular ABAP design using PERFORM subroutines. The overall flow is: Selection Screen → Data Retrieval (FORM f_get_data) → Field Catalog Build (FORM f_build_fieldcat) → Layout & Sort setup → ALV Display.

START-OF-SELECTION	Calls all processing FORMs in sequence
FORM f_get_data	SELECT from BSIK/BSAK; enrich with LFA1 vendor name
FORM f_build_fieldcat	Builds column definitions for ALV using DEFINE macro
FORM f_set_layout	Enables zebra stripes, column optimization, totals
FORM f_set_sort	Sorts by Company Code and Vendor with subtotals
FORM f_display_alv	Calls REUSE_ALV_GRID_DISPLAY to render the grid
FORM f_user_command	Handles double-click: navigates to FB03
FORM f_get_variant	F4 help for ALV display variants

4.2 Key Code Snippet – Data Retrieval

The data fetch combines open and cleared items via two separate SELECT statements, then enriches the result with vendor names using FOR ALL ENTRIES:

```
SELECT bukrs lifnr belnr gjahr bldat budat blart dmbtr waers zterm zfbdt FROM
bsik INTO TABLE lt_bsik
WHERE bukrs IN s_bukrs AND lifnr IN s_lifnr
AND bldat IN s_bldat AND gjahr IN s_gjahr AND
blart = p_blart.

SELECT lifnr name1 FROM lfa1 INTO TABLE lt_lfa1
FOR ALL ENTRIES IN gt_vendor_invoice WHERE
lifnr = gt_vendor_invoice-lifnr.
```

4.3 Screenshot Placeholders

After executing the program in SAP SE38 / SA38, paste your actual screenshots below. Replace the placeholder boxes with real screenshots from your SAP system.

5. Unique Points & Future Improvements

5.1 What Makes This Project Unique

#	Unique Point
1	Combines BSIK and BSAK in a single output — not available in any standard SAP report
2	Payment Status column ("Open"/"Cleared") gives instant visual classification
3	Double-click drill-down to FB03 enables seamless document investigation
4	DEFINE macro used for field catalog — reduces repetition and improves maintainability
5	FOR ALL ENTRIES used for efficient vendor name lookup — avoids nested SELECT loops
6	Subtotals at Company Code and Vendor level with Grand Total row

5.2 Future Improvements

Enhancement	Benefit
Email Notification	Auto-email overdue invoices to AP team using SAP Business Workpla
Overdue Highlighting	Color-code rows in red where invoice is open beyond payment due date
Ageing Analysis	Add 0-30, 31-60, 61-90 day buckets to classify outstanding invoices
Currency Conversion	Add group currency (GBP/USD) alongside local currency amounts
OData / Fiori Extension	Expose the report as a Fiori tile using SEGW for self-service access
BSEG Join	Join BSEG for GL account details to support reconciliation use cases

Note: This project was developed individually as a Capstone Project submission. All ABAP code is original. Replace all bracketed placeholders [YOUR NAME], [YOUR ROLL NUMBER], [YOUR BATCH/PROGRAM] with your actual details before submission.